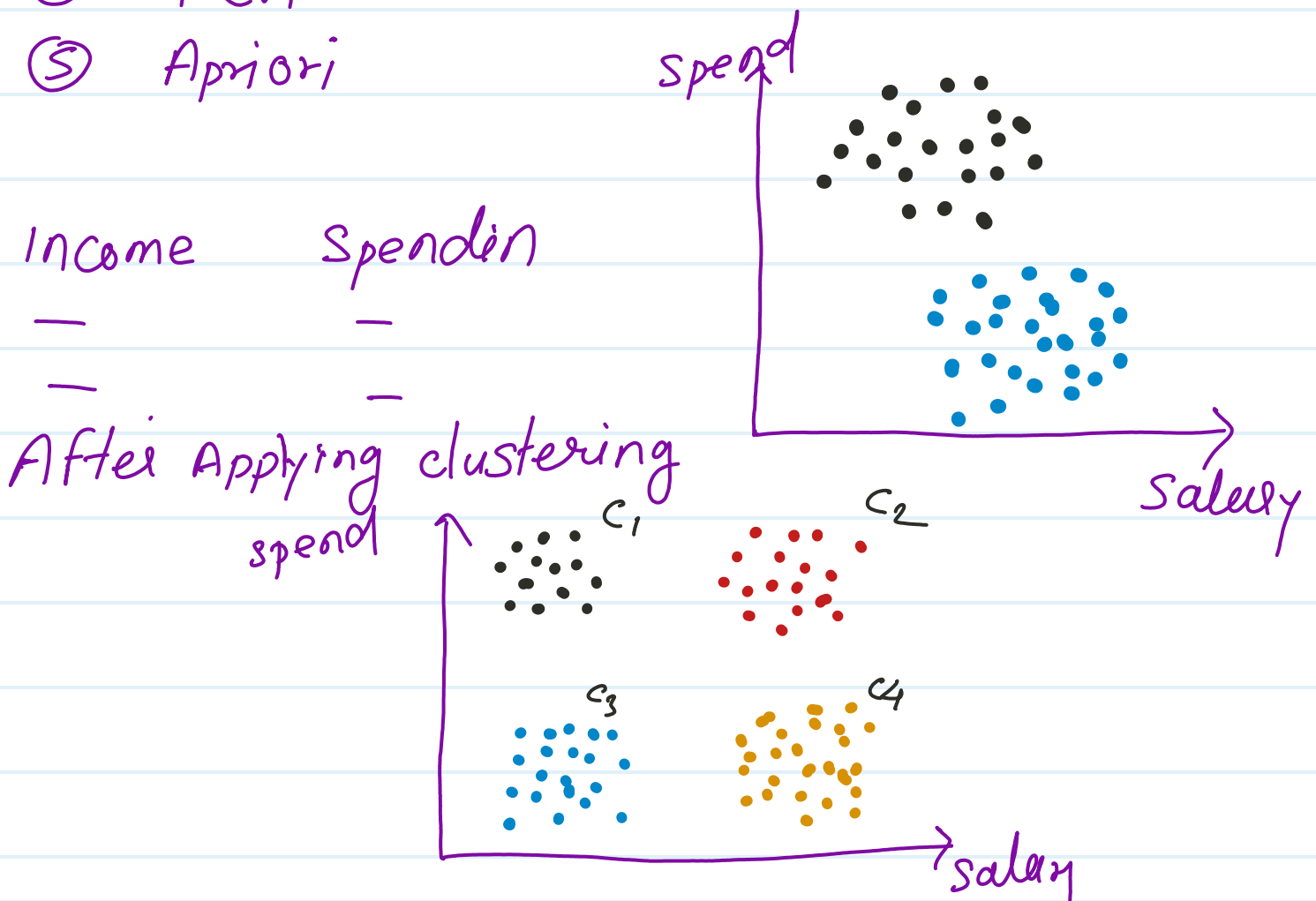


Unsupervised Learning

It does not required lab's data.

It can be work on structure and ~~Se~~ unstructure data.

- ① k-mean clustering
- ② Hierarchical clustering
- ③ DBSCAN clustering
- ④ PCA
- ⑤ Apriori



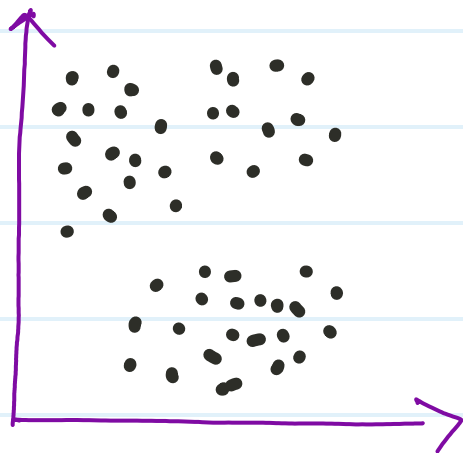
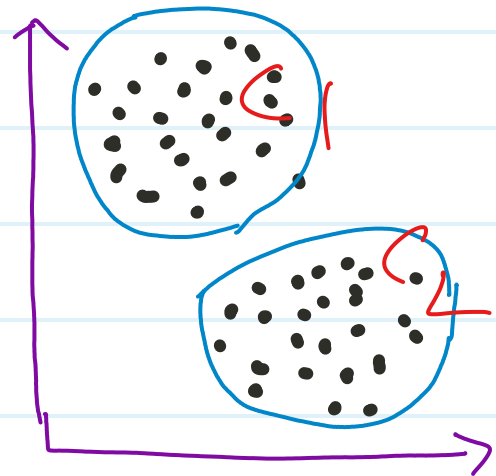
k-mean clustering

k-mean
++

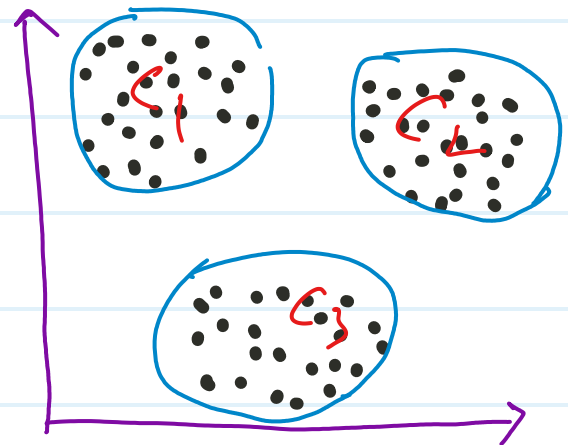
Geometric Intuition:-



Kmean $\Rightarrow$



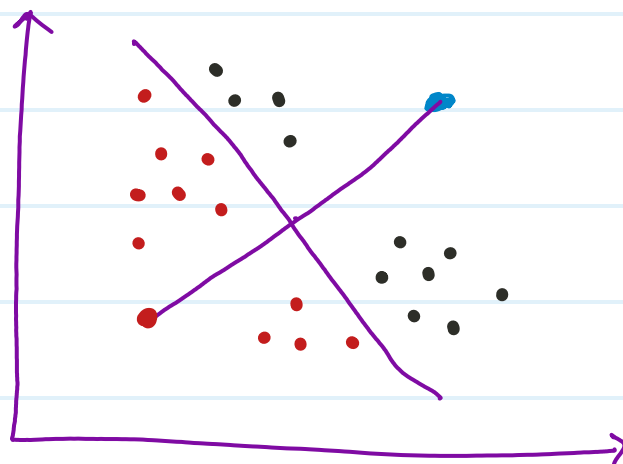
K-mean $\Rightarrow$



K =

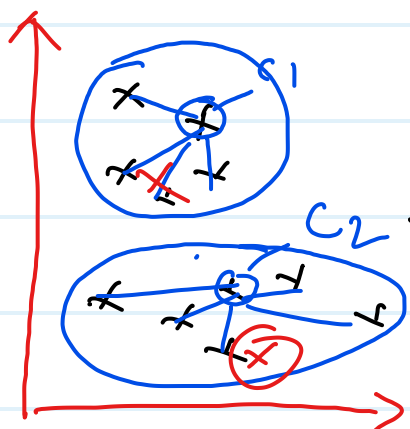
steps -

- ① Initialize some new centroid ^(K)
- ② point that are nearest to the centroid group them.
- ③ move centroid by calculating the mean of points

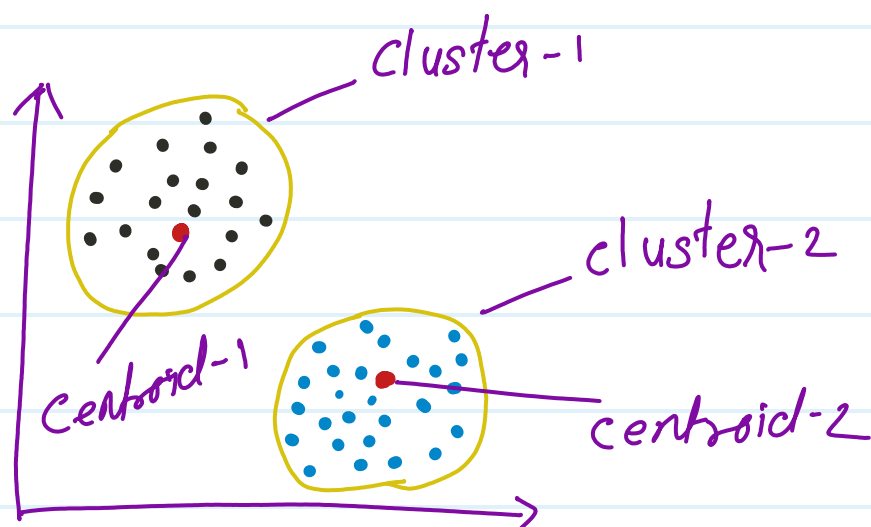


centroid point will shift toward. . . .

$k=2$

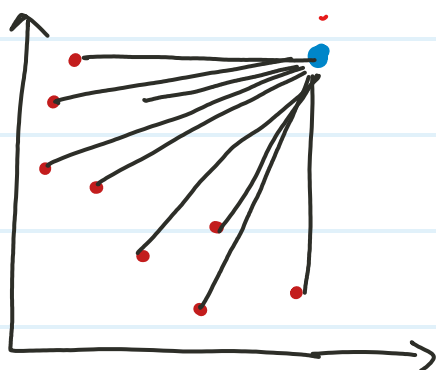


Final
output



How to select k-value?

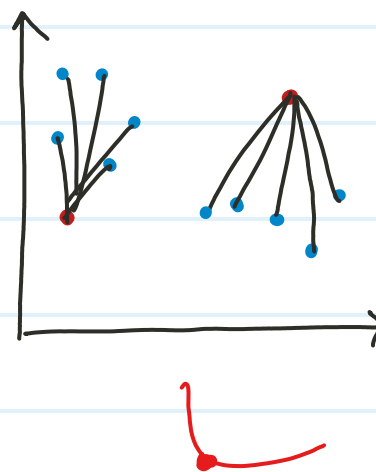
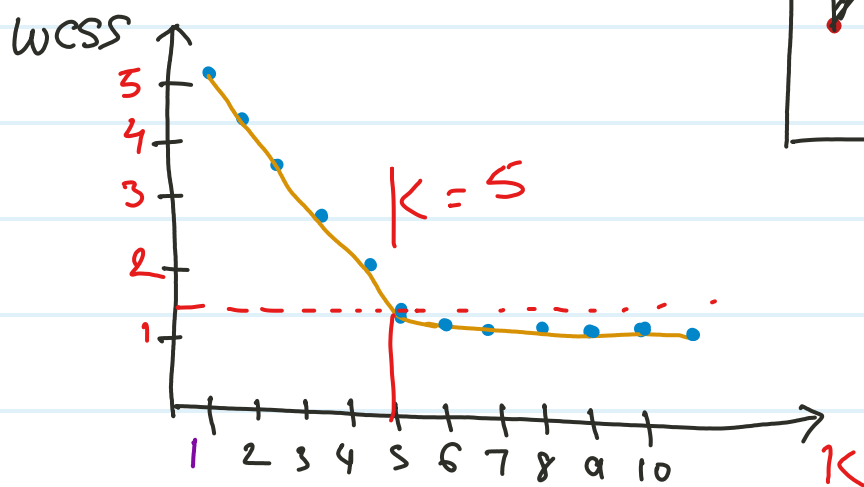
WCSS - within cluster sum of square



$$WCSS = \sum_{i=0}^k \left(\text{Distance b/w point to nearest centroid} \right)^2$$

suppose $k = 1 - 20$

* Elbo method -



when wcss calculate with $k=1$, wcss value is high, when we increase $k=2$, it will reduce, when $k=3$ it will reduce. when $k=3$ it will reduce and after

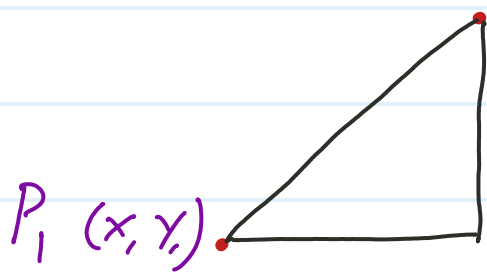
Some point it will be straight in graph it is called elbo method.

When curve become straight at that point we got k value. like above graph, 5 will be our k value.

To calculate distance we need.

Euclidean distance or Manhattan distanc.

$P_2 (x_2, y_2)$

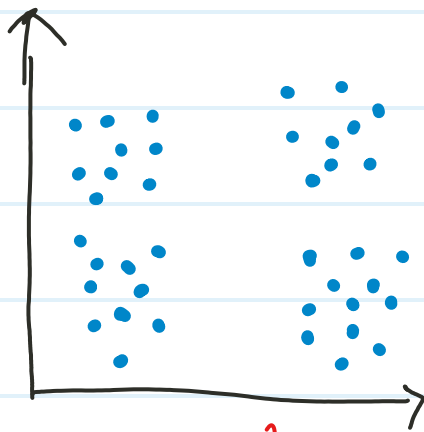


$$ED = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

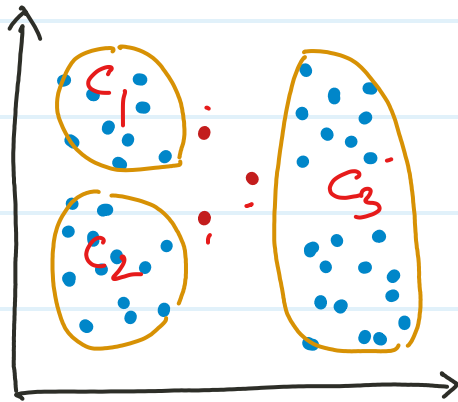
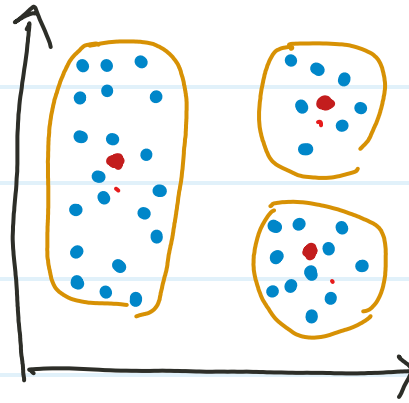
$$MD = |x_2 - x_1| + |y_2 - y_1|$$



Random Initialization trap (K-mean++)



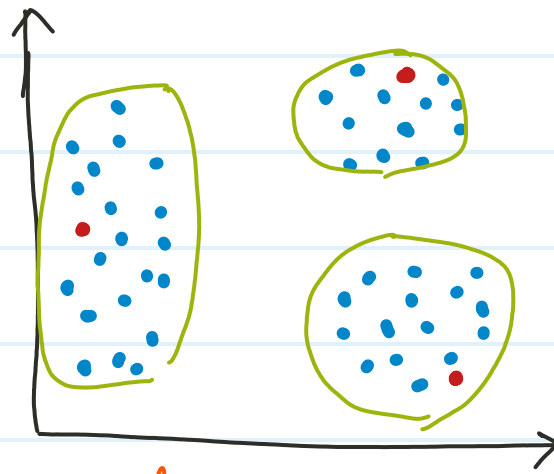
$K=3$



When centroid form very nearest to each other it stuck in Random trap.

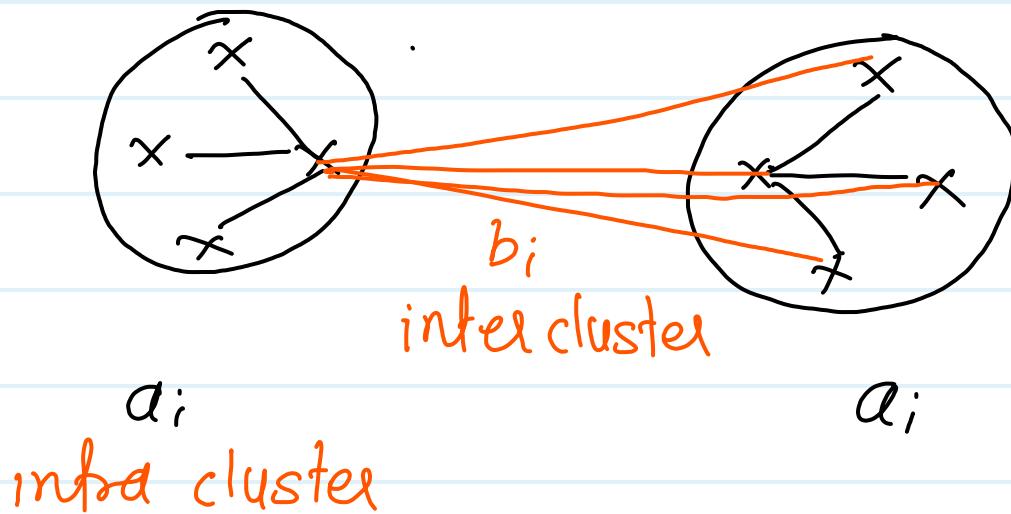
To handle this kind of trap we use K-mean++ method.

With help of it centroid form far from each other. and prevent from random initialization trap.



Evaluation matrix or performance matrix

① silhouette score -



$\Rightarrow b_i > a_i$ ✓ good

$b_i = a_i$ not good

$b_i < a_i$ very bad.